

BINF5501 Bioinformatics and Precision Medicine:

Interrogation of Variant Function Assignment

50% of final grade

Background:

Genetic variants can influence disease risk and response to therapy. Numerous experimental model systems have been developed that enable researchers to study the functional effects of genetic variants in controlled settings. The results of these experiments have been used to support assertions of variant pathogenicity in the clinic and to train artificial intelligence models that predict variant function.

Assignment objective:

You and your assigned research team have identified approximately 100 genetic variants localized to a particular gene or genomic locus. Although some of variants have been characterized using traditional methods, many of the variants you have identified are classified in ClinVar as variants of uncertain significance, or have no annotation.

Use your expert knowledge to develop a systematic experimental approach to study these variants. You may choose to work on any gene (except *TP53*, which we will discuss in class) and any disease or drug. The variants may be coding or non-coding and you may deploy molecular and/or phenotypic assays of your choosing.

How to complete the assignment:

This assignment has **3 Parts**:

Part 1: Pre-submission inquiry

Your team will be required to submit a *brief* (less than 200 word) pre-submission inquiry that nominates the gene and disease or drug. You must justify your selection of gene and disease or drug by citing at least two primary articles from reputable scientific journals and/or relevant variant annotation databases.

Part 1: Due **March 9, 2026 11:59PM** – Submit your Document in the “Submission” box in the Assignment #2 section on the Blackboard site.

Note: Ideally, each team would choose to study a different gene. Teams submitting duplicate inquiries will be informed and encouraged to modify their request.

Part 2: Proposal

Prepare a written proposal comprised of three sections.

The first section should provide the **Rationale** that links the gene of interest to the disorder or drug and describes the molecular consequences of known mutations. This should be an extension of your Pre-submission inquiry.

The second section should contain a detailed **Screening Plan** that describes the molecular and/or phenotypic assays that you would use to screen the variants in medium- to high-throughput. Be sure to comment on your choice of cell type(s), genetic constructs, data collection tools, and analytical techniques. Also describe how you would use positive and negative controls to benchmark your experiments and analyses.

The third section should comprise your **Validation Plan**, which may describe how you would test the function of your top scoring variants in low-throughput, high information-content assays and/or outline how you would integrate your screening data with existing databases to increase confidence in your findings.

You may choose to include additional subheadings in each section for clarity.

Quantitative and integrative analyses of published datasets or databases are highly encouraged.

Points will be awarded for novelty and feasibility.

The proposal should be between 1000 and 1500 words and may include up to 3 captioned figures or diagrams that do not contribute to the word count.

A separate **References** section (that has no limit on references and does not contribute to the word count) should be included at the end of the assignment. Primary research articles from reputable scientific journals should be cited in *Nature* format.

Part 2: Due **March 23, 2026 10 AM (BEFORE CLASS)** – Submit your Word Document in the “Submission” box in the Assignment #2 section on the Blackboard site.

Part 3: Presentation

Your team will also give a 25-minute PowerPoint presentation of your proposal in class, with 5 additional minutes allotted for questions and discussion.

Takes place **March 23, 2026 and March 30, 2026** – Be prepared to share your PowerPoint presentation.

Additional resources

ClinVar: <https://www.ncbi.nlm.nih.gov/clinvar/>

OMIM: <https://www.omim.org>

FDA pharmacogenomic table: <https://www.fda.gov/drugs/science-and-research-drugs/table-pharmacogenomic-biomarkers-drug-labeling>

gnomAD: <https://gnomad.broadinstitute.org>

cBioPortal: <https://www.cbioportal.org>

OncokB: <https://www.oncokb.org>

DepMap: <https://depmap.org/portal/>

EVE evolutionary model of variant effects: <https://evemodel.org>

ESM Variants: https://huggingface.co/spaces/ntranoslab/esm_variants/

AlphaMissense: <https://zenodo.org/record/8360242>

Evaluation Rubric:

Multiplier	Criteria				
	Pre-submission Inquiry clearly articulates the association between target and disease				
	Very Poor	Poor	Adequate	Good	Very Good
10	0	0.25	0.5	0.75	1
	The rationale to select the target is justified using scientific literature.				
2	0	0.25	0.5	0.75	1
	The rationale to select the target is justified using database analysis.				
2	0	0.25	0.5	0.75	1
	The scientific question is novel.				
2	0	0.25	0.5	0.75	1
	The scientific question is focused and specific.				
5	0	0.25	0.5	0.75	1
	The scientific question is likely to advance the field.				
2	0	0.25	0.5	0.75	1
	The rationale to select controls is justified.				
5	0	0.25	0.5	0.75	1
	The rationale to select a model system is justified.				
5	0	0.25	0.5	0.75	1
	The proposed approaches are described accurately, and in appropriate detail.				
5	0	0.25	0.5	0.75	1
	The proposed approaches are likely to answer the scientific question.				
5	0	0.25	0.5	0.75	1
	The proposed approaches are efficient.				
5	0	0.25	0.5	0.75	1
	The proposed approaches may yield insights beyond the proposed scientific question.				
2	0	0.25	0.5	0.75	1
	The proposed approaches describe how the data will be integrated with specific existing databases to corroborate the findings.				
5	0	0.25	0.5	0.75	1
	The proposed approaches are feasible in a well-funded lab in a reasonable time frame.				
2	0	0.25	0.5	0.75	1
	The writing style and formatting are consistent throughout.				
2	0	0.25	0.5	0.75	1
	The logic is clear and consistent throughout.				
2	0	0.25	0.5	0.75	1
	References are cited in line with text and on slides, using APA or Nature format and included in a bibliography				
5	0	0.25	0.5	0.75	1
	All figures and tables are high-resolution.				
2	0	0.25	0.5	0.75	1
	All axes, tables, and text are clear.				
1	0	0.25	0.5	0.75	1

	Figures and tables are well-organized and visually-appealing.				
5	0	0.25	0.5	0.75	1
	All figures have captions with appropriate detail.				
2	0	0.25	0.5	0.75	1
	Statistical and computational analyses are described in appropriate detail.				
5	0	0.25	0.5	0.75	1
	Description of expected or observed results incorporates directional language (higher, lower, no difference etc.)				
2	0	0.25	0.5	0.75	1
	The potential implications of the study were described.				
5	0	0.25	0.5	0.75	1
	The length of the report was appropriate.				
5	0	0.25	0.5	0.75	1
	All audience and instructor questions were thoughtfully considered.				
1	0	0.25	0.5	0.75	1
	The answers to audience and instructor questions were correct and/or instructive.				
1	0	0.25	0.5	0.75	1
	All group members were present at all presentations.				
1	0	0.25	0.5	0.75	1
	All group members presented equally.				
1	0	0.25	0.5	0.75	1
	All group members used a professional tone.				
1	0	0.25	0.5	0.75	1
	All group members were respectful.				
1	0	0.25	0.5	0.75	1
	All group members asked at least one question of a presenting group.				
1	0	0.25	0.5	0.75	1
	Total				
100					
	The instructor believes this report to be inappropriately influenced by Artificial Intelligence (subtracted from Total)				
-100	-100	-75	-50	-25	-0
	Peer evaluation score (multiplier on Total)				
1	0	0.25	0.5	0.75	1

Peer Evaluation Criteria

Multiplier	Criteria										
	This group member was present at all group discussion sessions.										
	Did not meet expectations Met expectations										
0.25	0	1	2	3	4	5	6	7	8	9	10
	This group member contributed equally to the project between discussion sessions.										
0.25	0	1	2	3	4	5	6	7	8	9	10
	This group member submitted group tasks in a timely manner.										
0.25	0	1	2	3	4	5	6	7	8	9	10
	This group member reviewed and provided feedback on the final product.										
0.25	0	1	2	3	4	5	6	7	8	9	10